THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 65 degrees	i = 90 - 25 = 65.
2	**(b)** 20 cm	R = 2f.
3	**(b)** principal focus	Focus.
4	**(b)** 40 degrees	i + r = 80.
5	**(d)** real, inverted, same size	At C.
6	**(b)** virtual, erect and smaller	Always.
7	**(a)** 4 m/s	Relative speed.
8	**(a)** virtual, erect and enlarged	Magnifying mirror.
9	**(a)** at the focus	Parallel rays.
10	**(c)** 2 m	2 x 1.0.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	i = 60 degrees (1); Total angle = 120 degrees (1).	1+1
12	Convex mirror diverges rays (1); A virtual, diminished image is formed (1).	1+1
13	Rays meet at the focus (1); An inverted real image forms (1).	1+1
14	Gives an enlarged, erect image when close (1); Shows teeth in detail (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New i = 50 degrees (or 30) (1); Reflected ray turns by twice 10 (1); = 20 degrees (1).	1+1+1
16	Wide field of view (1); Image erect (1); Objects appear smaller and farther (1).	1+1+1
17	Concave: converges at the focus (1); Convex: diverges, appears to come from the focus behind the mirror (1); Different effects of curvature (1).	1+1+1
	OR: Polished surface gives regular reflection (1); Rough surface scatters (1); Image forms only in regular reflection (1).	1+1+1
18	Reflected at 30 degrees from the first mirror (1); Strikes second at 60 degrees to its normal (1); Emergent ray is antiparallel to the incident ray (1).	1+1+1
	OR: Lens forms an inverted real image (1); Brain interprets it as upright (1); Learned by the brain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mirror, white paper, pins, protractor (1); Draw normal and incident ray (1); Fix pins on reflected ray (1); Measure i and r (1); i = r, all in one plane (1).	1+1+1+1+1
	OR: Minimum height = 0.9 m (1); Top edge at half the distance of head and eyes (1); Bottom edge at half the distance of feet and eyes (1); Independent of distance from mirror (1); Mirror length is half (1).	1+1+1+1+1
20	n = 360/60 - 1 = 5 (1); Images by repeated reflection (1); At 90 degrees n = 3 (1); Smaller angle gives more images (1); Parallel mirrors give infinite images (1).	1+1+1+1+1
	OR: Eye lens is convex (1); Myopia: image in front of retina (1); Corrected by a concave lens (1); Hypermetropia: image behind retina (1); Corrected by a convex lens (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** principal focus	Focus.
2	**(c)** convex lens of small focal length	Closer to object than f.
3	**(c)** half the height of the person	Property of plane mirrors.
4	**(a)** parallel to the principal axis	Reverse of parallel ray.
5	**(b)** erect and enlarged	Object within f.
6	**(c)** 7	360/45 - 1.
7	**(c)** retraces its path	Along the normal.
8	**(c)** inverted	Object beyond f.
9	**(a)** infinite	Repeated reflections.
10	**(d)** virtual, erect and diminished	Always smaller.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	360/72 - 1 (1); = 4 images (1).	1+1
12	Real: rays actually meet (1); Virtual: rays appear to meet (1).	1+1
13	Converge sunlight to a focus (1); Heating there (1).	1+1
14	It is laterally inverted (1); Reads correctly in the rear-view mirror (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Object between pole and focus (1); Reflected rays diverge, appear to meet behind (1); Image is virtual, erect, enlarged (1).	1+1+1
16	Reflectors send light forward (1); Increase brightness (1); Beam does not spread (1).	1+1+1
17	Draw a normal on paper under a mirror (1); Fix pins along the incident and reflected rays (1); Measure i and r (1).	1+1+1
	OR: Ray parallel to the axis passes through the focus (1); Ray through the optical centre goes undeviated (1); Ray through the focus emerges parallel (1).	1+1+1
18	Place the candle beyond the focal length (1); Screen on the other side (1); Real, inverted image (1).	1+1+1
	OR: The objects are very far (1); Rays are nearly parallel (1); They converge at the focus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Beyond C: real, inverted, smaller (1); At C: same size (1); Between C and F: enlarged (1); At F: at infinity (1); Between F and P: virtual, erect, larger (1).	1+1+1+1+1
	OR: Light passes through a small hole (1); Inverted image on screen (1); Dim image (1); Lens camera gives a bright image (1); Focusing is needed with lens (1).	1+1+1+1+1
20	Beyond 2F: real, inverted, smaller (1); Between F and 2F: real, inverted, larger (1); Within F: virtual, erect, enlarged (1); Rules of three rays (1); Image on the other side for real (1).	1+1+1+1+1
	OR: Magnification = image size / object size (1); Magnifier gives enlarged image (1); Change distance between lens and object (1); Observe clarity (1); Best when object is within f (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** convex	Converges.
2	**(c)** 45 degrees	i + r = 90.
3	**(a)** concave	Depends on distance.
4	**(a)** diverge	Diverging lens.
5	**(c)** 8:15	12:00 - 3:45 = 8:15.
6	**(d)** infinity	Parallel rays stay parallel.
7	**(c)** concave	Concave.
8	**(a)** A	A is symmetric (mirror line vertical).
9	**(a)** concave mirror	Close object.
10	**(a)** convex lens	Forms real image on retina.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gives regular reflection (1); Smooth surface reflects almost all light (1).	1+1
12	New distance 25 cm (1); Object-image distance = 50 cm (1).	1+1
13	Central point of the lens (1); Rays through it go undeviated (1).	1+1
14	Camera and projector: convex (1); Myopia: concave (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Convex is thick at the centre; concave is thin (1); Convex converges; concave diverges (1); Convex: magnifier, camera; concave: spectacles for myopia (1).	1+1+1
16	Plane has a flat surface (1); Plane always gives virtual erect image; spherical depends (1); Spherical mirrors have a focus (1).	1+1+1
17	The image depends on light reflected from the object (1); A mirror reflects regularly (1); In the dark nothing reaches it (1).	1+1+1
	OR: Place object within the focal length (1); Virtual, erect, enlarged image forms (1); Used by watchmakers (1).	1+1+1
18	f = 12 cm (1); At the principal focus (1); 12 cm from the pole (1).	1+1+1
	OR: Wide field of view (1); Diminished image of many objects (1); Helps security (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Plane: dressing (1); Concave mirror: shaving (1); Convex mirror: rear-view (1); Convex lens: camera, magnifier (1); Concave lens: spectacles for myopia (1).	1+1+1+1+1
	OR: Parallel rays converge to a point (1); Energy concentrated (1); Temperature rises (1); Paper ignites at the focus (1); Distance equals focal length (1).	1+1+1+1+1
20	Periscope: two parallel mirrors at 45 degrees (1); Light reflects twice (1); Kaleidoscope: three mirrors at 60 degrees (1); Multiple reflections (1); Uses: submarines, toys (1).	1+1+1+1+1
	OR: Reflection reverses the depth direction (1); Left becomes right (1); F appears reversed (1); 3 appears reversed (1); Up-down is unchanged (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.